

LAB EXPERIMENT – 1

Versioning and Tracking Machine Learning Models using MLflow

Course Outcome: CO1

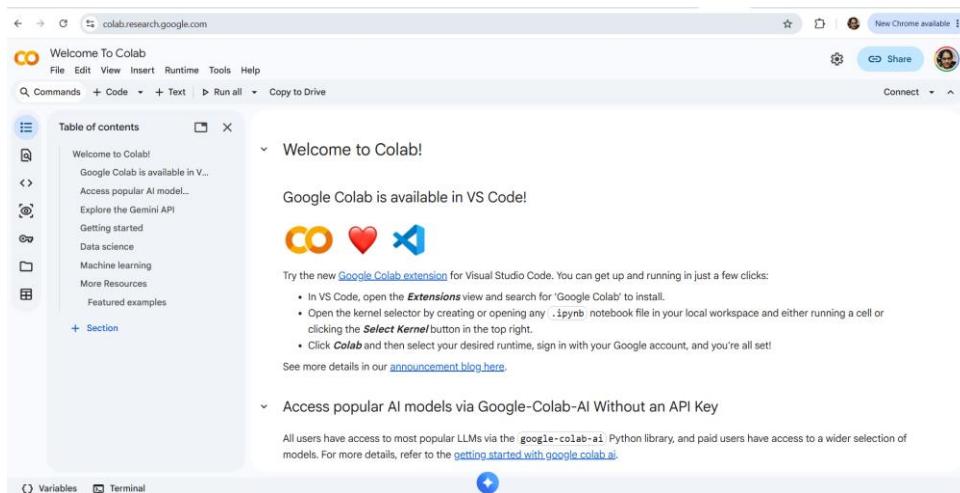
Aim

To implement experiment tracking, parameter logging, metric logging, and model versioning using MLflow.

Step 1: Open Google Colab

Open your browser and visit:

<https://colab.research.google.com>



Step 2: Create a New Notebook

Click

File → New Notebook

Rename the notebook as

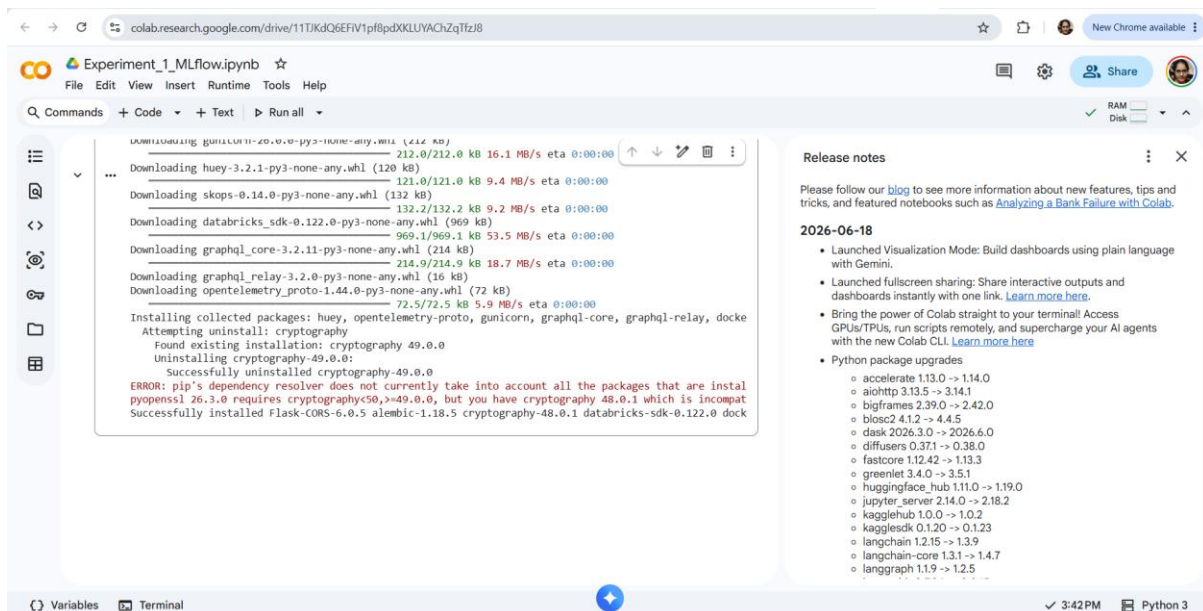
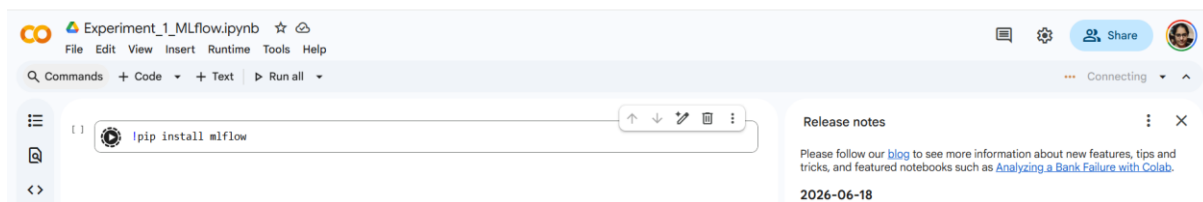
Experiment_1_MLflow.ipynb

Step 3: Install MLflow

Create a new code cell and execute

!pip install mlflow

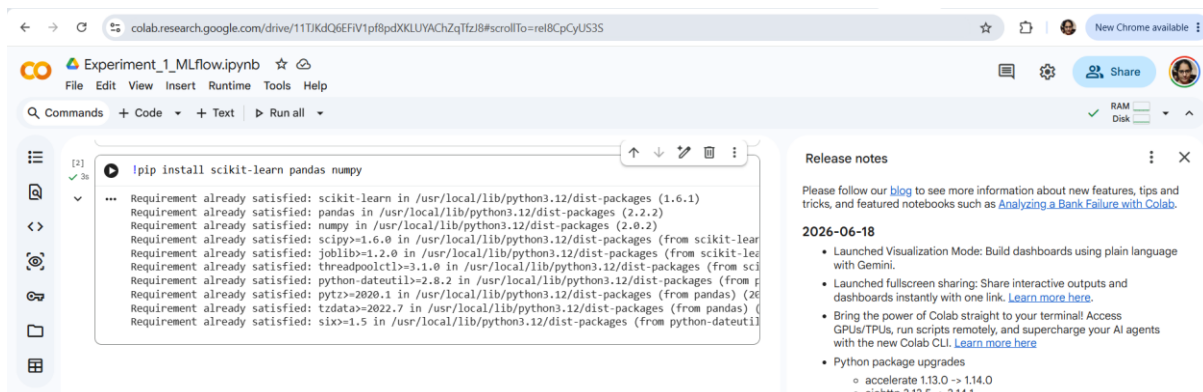
Wait until installation completes.



Step 4: Install Required Libraries

Create another cell.

`!pip install scikit-learn pandas numpy`



Step 5: Import Libraries

Create another code cell.

`import mlflow`

`import mlflow.sklearn`

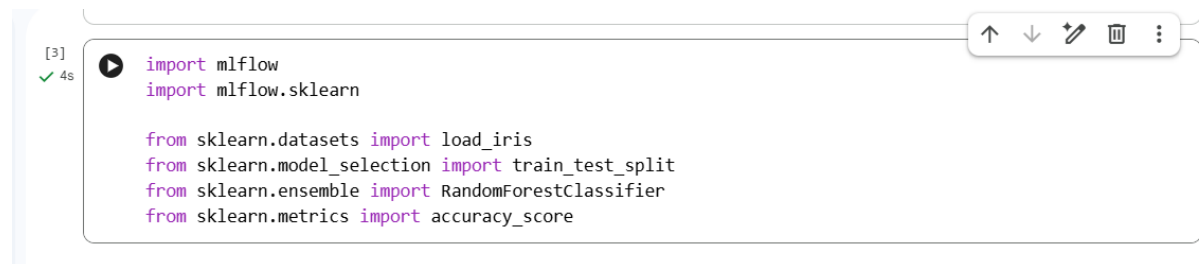
```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score
```

Run the cell.

A screenshot of a Jupyter Notebook cell. On the left, there is a status bar showing '[3]' and a green checkmark with '4s'. The cell contains Python code for importing mlflow, sklearn, and various sklearn modules. On the right, there is a toolbar with icons for undo, redo, insert, delete, and a menu.

```
[3] ✓ 4s import mlflow
import mlflow.sklearn

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

Step 6: Load Dataset

Create a new cell.

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

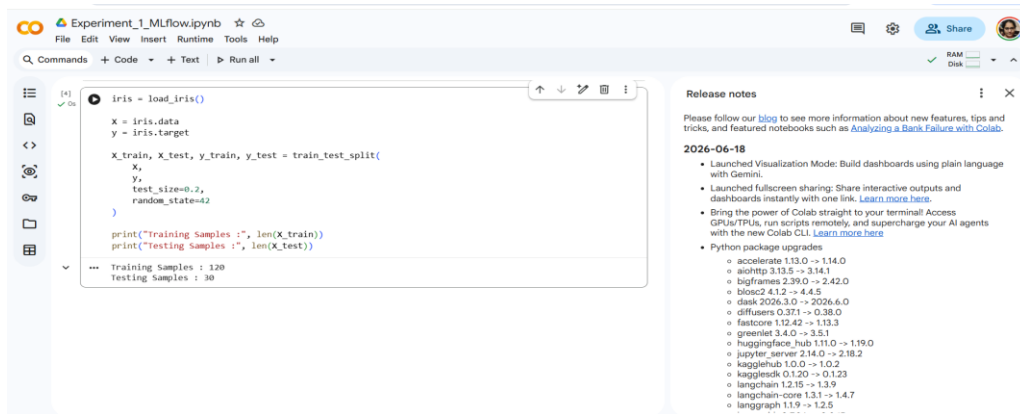
```
print("Training Samples :", len(X_train))
```

```
print("Testing Samples :", len(X_test))
```

Expected Output

Training Samples : 120

Testing Samples : 30

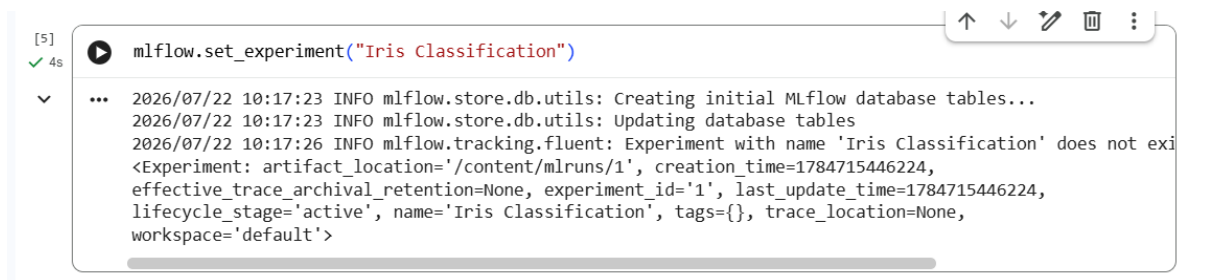


Step 7: Create an MLflow Experiment

Create another cell.

```
mlflow.set_experiment("Iris Classification")
```

Run the cell.



Step 8: Train the Machine Learning Model

Create another code cell.

with mlflow.start_run():

```
n_estimators = 100
```

```
max_depth = 5
```

```
model = RandomForestClassifier(
    n_estimators=n_estimators,
    max_depth=max_depth,
    random_state=42
)
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, predictions)
```

```
mlflow.log_param("Number of Trees", n_estimators)
```

```
mlflow.log_param("Maximum Depth", max_depth)
```

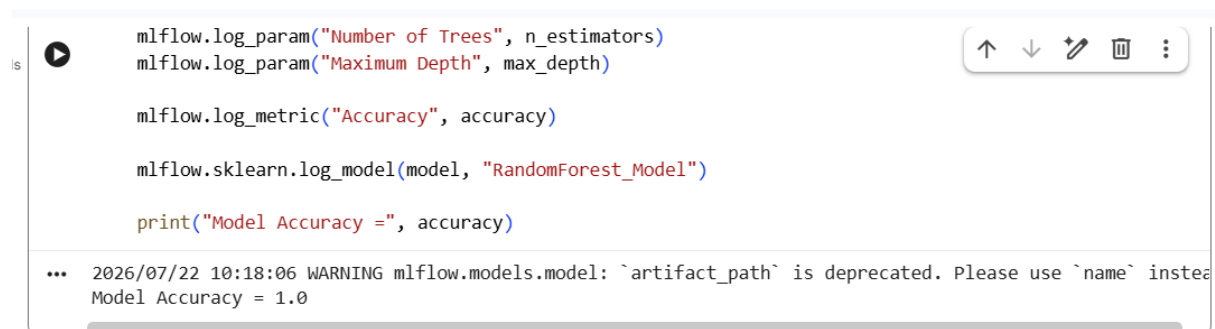
```
mlflow.log_metric("Accuracy", accuracy)
```

```
mlflow.sklearn.log_model(model, "RandomForest_Model")
```

```
print("Model Accuracy =", accuracy)
```

Expected Output

Model Accuracy = 1.0



```
mlflow.log_param("Number of Trees", n_estimators)
mlflow.log_param("Maximum Depth", max_depth)

mlflow.log_metric("Accuracy", accuracy)

mlflow.sklearn.log_model(model, "RandomForest_Model")

print("Model Accuracy =", accuracy)
```

... 2026/07/22 10:18:06 WARNING mlflow.models.model: `artifact\_path` is deprecated. Please use `name` instead
Model Accuracy = 1.0

Step 9: Display Experiment Details

Create another cell.

```
print("Experiment Completed Successfully")
```

Output

Experiment Completed Successfully

Step 10: Save Notebook

Click

File↓Save

The notebook is automatically saved to Google Drive.